Chapter - 6

Discrete least square Method

let we have a matrix,

$5x_1 + 2x_2 + 3x_3 = 7$

$2x_1 + 7x_2 + 8x_3 = 5$

$3x_1 + 9x_2 + 2x_3 = 6$

So,

$$\begin{bmatrix} 5 & 2 & 3 \\ 2 & 7 & 8 \\ 3 & 9 & 2 \end{bmatrix}$$

This is $[3\times3]$ $[m\times n]$ square Matrix. Which is solvable by Gaussian & LU Decomposition method

Now, if we add another equation $(4x_1 + 2x + 5x_3 = 10)$ The matrix won't be square matrix anymore. So it won't be solvable using these techniques.

$$\begin{bmatrix} 5 & 2 & 3 \\ 2 & 7 & 8 \\ 3 & 9 & 2 \\ 4 & 2 & 5 \end{bmatrix}$$

row × col

$[4\times3]$

$[m\times n]$; $m > n$

This is known as Overdetermined System.

Cause row ≠ col, Not an square Matrix.

vars ≠ # eqn

Orthonormality

Orthogonal + Normality.

Row Matrix → $x = [x_0 \; x_1 \; x_2 \; x_3]$.

↳ $[1 \times 4]$

Column Matrix → $y = \begin{bmatrix} y_0 \\ y_1 \\ y_2 \\ y_3 \end{bmatrix}$ $[4 \times 1]$

Transposing a matrix means, Rows become cols and vise versa.

$x = [x_0 \; x_1 \; x_2 \; x_3] \rightarrow [1 \times 4]$ (RE)

$x^T = \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{bmatrix} \rightarrow [4 \times 1]$ (RC)

$y = \begin{bmatrix} 5 & 4 & 3 \\ 1 & 0 & 2 \end{bmatrix}$; $y^T = \begin{bmatrix} 5 & 1 \\ 4 & 0 \\ 3 & 2 \end{bmatrix}$

$[2 \times 3]$ $[3 \times 2]$

So, we can say by dimension, $Q_{(m \times n)}$ হবে $Q^T_{(n \times m)}$ হবে।

By Rules of Matrix multiplicataion,

$$Q^T_{(n \times m)} \; Q_{(m \times n)} = I_{(n \times n)}$$ which is a square matrix

[~~A (R C) B (R C) = C (R1 C2)~~]

A B = AB
(m) × n n × (p) m × p
↓ ↓
Equal
R1 R2

Orthogonality

A matrix is orthogonal if $x^T y = 0$

Here, $\theta = 90°$; $x^T y = |x| \, |y| \cos 90°$

$x^T y = 0$ // $x^T y = 0$, $y^T z = 0$, $x^T z = 0$

Normality

$x^T x = 1$; $y^T y = 1$

A matrix × Transposed matrix = 1

Check if the following set is orthonormal or not.

$$S = \left\{ \frac{1}{\sqrt{5}}(2,1)^T, \frac{1}{\sqrt{5}}(1,-2)^T \right\}$$

Soln

x y

1st checking Normality

Let, $x^T = \frac{1}{\sqrt{5}}(2,1)^T$; $y^T = \frac{1}{\sqrt{5}}(1,-2)^T$

Now, $x^T x = 1$

So, $x^T x = \frac{1}{\sqrt{5}}(2,1) \times \frac{1}{\sqrt{5}} \begin{pmatrix} 2 \\ 1 \end{pmatrix}$

$= \frac{1}{5}(2 \cdot 1) \cdot \begin{pmatrix} 2 \\ 1 \end{pmatrix}$

$= \frac{1}{5} \times 5 = 1$

So, $x^T x = 1$

Now, for $y^T y = \frac{1}{\sqrt{5}}(1 \ -2) \times \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ -2 \end{pmatrix}$

$= \frac{1}{5} \times 5 = 1$

So, $y^T y = 1$

As $x^T x = 1$ and $y^T y = 1$; so this matrix is Normal On this set has a normal property.

Now checking Orthogonality

if, $x^T y = 0$

Here, $x^T = \frac{1}{\sqrt{5}} (2 \; 1)$ $\quad y = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ -2 \end{pmatrix}$

So, $x^T y = \frac{1}{\sqrt{5}} (2 \; 1) \times \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ -2 \end{pmatrix}$

$= \frac{1}{5} (0) = 0$

So, it is Orthogonal too

So, the Vector set is Orthonormal

Kronecker Delta

$$\delta_{ij} = \begin{cases} 1 & ; i = j \\ 0 & ; i \neq j \end{cases}$$

(u to u when $i = j$)

- $u^T u$ [Normal]
- $u^T v$ [orthogonal]
- [orthonormal]

If given Matrix Not a square matrix then it can't be solved using Gaussian or LU decomposition. That's why we introduce Least Square Method.

$$\underset{[n \times n]}{Ax} = b$$

$$A^T A x = A^T b$$

$$[n \times m]\ [m \times n] = [n \times n]$$

Ex

$f(-3)=0$
$f(0)=0$
$f(6)=2$

Normally the polynomial would be,

$P_2(x) = a_0 + a_1 x + a_2 x^2$

But as we are aiming for an Overdetermined system, the polynomial will be 1 degree shorter. So, $P_1(x) = a_0 + a_1 x$

So, $P_1(x) = a_0 + a_1 x$

$\Rightarrow P_1(-3) \Rightarrow a_0 - 3a_1 = 0$

$\Rightarrow P_1(0) \Rightarrow a_0 = 0$

$\Rightarrow P_1(6) \Rightarrow a_0 + 6a_1 = 2$

As # eqn > # var
So, it is an Overdetermined System.

So, the equations are,

$$a_0 - 3a_1 = 0$$
$$a_0 = 0$$
$$a_0 + 6a_1 = 2$$

So, we got the equations.

$$a_0 - 3a_1 = 0$$
$$a_0 = 0$$
$$a_0 + 6a_1 = 2$$

Now converting into orthogonal Matrix.

$$\begin{bmatrix} 1 & -3 \\ 1 & 0 \\ 1 & 6 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$$

A $\quad$ x $\quad$ b

So, $Ax = b$

$\Rightarrow A^T A x = A^T b$

So, Now,

$$A^T A = \underset{(2\times 3)}{\begin{bmatrix} 1 & 1 & 1 \\ -3 & 0 & 6 \end{bmatrix}} \underset{(3\times 2)}{\begin{bmatrix} 1 & -3 \\ 1 & 0 \\ 1 & 6 \end{bmatrix}}$$

$$\therefore = \underset{(2\times 2)}{\begin{bmatrix} 3 & 3 \\ 3 & 45 \end{bmatrix}}$$

Now,

$$A^T b = \underset{(2\times3)}{\begin{bmatrix} 1 & 1 & 1 \\ -3 & 0 & 6 \end{bmatrix}} \quad \underset{(3\times1)}{\begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}}$$

$$= \begin{bmatrix} 2 \\ 12 \end{bmatrix}$$

So,

$$A^T A x = A^T b$$

$$\Rightarrow \begin{bmatrix} 3 & 3 \\ 3 & 45 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 2 \\ 12 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3 & 45 \end{bmatrix}^{-1} \begin{bmatrix} 2 \\ 12 \end{bmatrix}$$

$$= \frac{1}{(3\times45) - (3\times3)} \begin{bmatrix} 45 & -3 \\ -3 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 12 \end{bmatrix}$$

$$= \begin{bmatrix} 0.35714 & 0.02380 \\ 0.02380 & 0.02380 \end{bmatrix} \begin{bmatrix} 2 \\ 12 \end{bmatrix}$$

$$= \begin{bmatrix} 3/7 \\ 5/21 \end{bmatrix}$$

Solved Using Vandermonde Matrix

So, $a_0 = \frac{3}{7}$; $a_1 = \frac{5}{21}$

So, $P_1(x) = \frac{3}{7} + \frac{5}{21}x$

$f(2) = 3$ | So,

$f(3) = 5$

$f(5) = 12$

$f(6) = 15$

$P_3(x) \rightarrow a_0 + a_1 x + a_2 x^2 = 0$

$P_3(2) \rightarrow a_0 + 2a_1 + 4a_2 = 3$

$P_3(3) \rightarrow a_0 + 3a_1 + 9a_2 = 5$

$P_3(5) \rightarrow a_0 + 5a_1 + 25a_2 = 12$

$f(2) = 3$; $f(3) = 5$; $f(5) = 12$; $f(6) = 15$

Soln We know,

$P_2(x) \rightarrow a_0 x + a_1 x + a_2 x^2 = 0$

$P_2(2) \rightarrow a_0 + 2a_1 + 4a_2 = 3$

$P_2(3) \rightarrow a_0 + 3a_1 + 9a_2 = 5$

$P_2(5) \rightarrow a_0 + 5a_1 + 25a_2 = 12$

$P_2(6) \rightarrow a_0 + 6a_1 + 36a_2 = 15$

Now converting into orthogonal matrix,

$$\begin{bmatrix} 1 & 2 & 4 \\ 1 & 3 & 9 \\ 1 & 5 & 25 \\ 1 & 6 & 36 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \\ 12 \\ 13 \end{bmatrix}$$

A $\quad$ x $\quad$ b

So, $Ax = b$

$\therefore A^T A x = A^T b$

Now, $A^T A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 3 & 5 & 6 \\ 4 & 9 & 25 & 36 \end{bmatrix}_{3\times4} \begin{bmatrix} 1 & 2 & 4 \\ 1 & 3 & 9 \\ 1 & 5 & 25 \\ 1 & 6 & 36 \end{bmatrix}_{4\times3}$

$$= \begin{bmatrix} 4 & 16 & 74 \\ 16 & 74 & 376 \\ 74 & 376 & 2018 \end{bmatrix}_{3\times3}$$

Now, $A^T b = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 3 & 5 & 6 \\ 4 & 9 & 25 & 36 \end{bmatrix}_{3\times4} \begin{bmatrix} 3 \\ 5 \\ 12 \\ 15 \end{bmatrix}_{4\times1}$

$$= \begin{bmatrix} 35 \\ 171 \\ [illegible] \end{bmatrix}_{3\times1}$$

Solving Using Gaussian,

$$\boxed{A^T A x = A^T b}$$

$$\left[\begin{array}{ccc|c} 4 & 16 & 74 & 35 \\ 16 & 74 & 376 & 171 \\ 74 & 376 & 2018 & 897 \end{array}\right]$$

$R_2' = R_2 - \left(\frac{4}{1}\right) R_1$

$R_3' = R_3 - \left(\frac{74}{4}\right) R_1$

$$= \left[\begin{array}{ccc|c} 4 & 16 & 74 & 35 \\ 0 & 10 & 80 & 31 \\ 0 & 80 & 649 & 245.5 \end{array}\right]$$

$R_3' = R_3 - \left(\frac{80}{10}\right) R_3$

$$= \left[\begin{array}{ccc|c} 4 & 16 & 74 & 35 \\ 0 & 10 & 80 & 31 \\ 0 & 0 & 9 & 1.5 \end{array}\right]$$

So, Now,

$9x_2 = 1.5$

So, $x_2 = \frac{1}{6}$

$10x_1 + 80x_2 = 31$

$\Rightarrow 80x_2 = 31 - 10x_1$

$\therefore x_3 =$

$10x_1 + 80x_2 = 31$

$\Rightarrow x_1 = (31 - 80x_2)/10$

$\therefore x_1 = 53/30$

$4x_0 + 16x_1 + 74x_2 = 35$

$\Rightarrow 4x_0 = -7/5$

QR Decomposition

$$A_{m \times n} = Q R$$

- Q = Special Matrix, All the Col's are orthonormal
- R = Upper Triangular Matrix

Orthonormal Property ; Gram Schmidt process

Let we have a Co-efficient Matrix,

$$A = \begin{bmatrix} 3 & 1 \\ 6 & 2 \\ 0 & 2 \end{bmatrix} \quad \begin{matrix} u_1 & u_2 \end{matrix}$$

Using Gram Schmidt process formula,

$$P_k = u_k - \sum_{i=1}^{k-1} \left(u_k^T q_i \right) q_i$$

→ Scaler Dot Product

or, $$P_k = u_k - \sum_{i=1}^{k-1} \frac{u_k \cdot P_i}{P_i \cdot P_i} P_i$$

$$q_k = \frac{P_k}{|P_k|}$$

$$\# \quad A = \begin{matrix} u_1 & u_2 \end{matrix} \begin{bmatrix} 3 & 1 \\ 6 & 2 \\ 0 & 2 \end{bmatrix}$$

Now, let, $u_1 = \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix}$ and $u_2 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$

for step-1 ; $k = 1$

$$P_k = u_k - \sum_{i=1}^{k-1} \frac{u_k \cdot P_i}{P_i \cdot P_i} P_i$$

So, $P_1 = u_1 = \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix}$

$$q_1 = \frac{P_1}{|P_1|} = \frac{1}{\sqrt{3^2+6^2+0^2}} \times \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix}$$

$$= \frac{1}{\sqrt{45}} \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix}$$

So, $P_1 = \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix}$; $q_1 = \frac{1}{\sqrt{45}} \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix}$

For step-2 ; k=2

Process-1

$$p_k = u_k - \sum_{j=1}^{k-1} (u_k^T q_i) q_i$$

So, $p_2 = u_2 - (u_2^T q_1) q_1$

$$= \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} - \left(\begin{bmatrix} 1 & 2 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix} \frac{1}{\sqrt{45}} \right) q_1$$

$$= \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} - \left(15 \times \frac{1}{\sqrt{45}} \right) \left[\begin{pmatrix} 3 \\ 6 \\ 0 \end{pmatrix} \frac{1}{\sqrt{45}} \right]$$

$$= \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix} - \left[\frac{1}{3} \begin{pmatrix} 3 \\ 6 \\ 0 \end{pmatrix} \right]$$

$$= \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} - \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$$

$$u_2 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \qquad P_1 = \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix}$$

Process-2

$$P_k = u_k - \sum_{i=1}^{k-1} \frac{u_k \cdot P_i}{P_i \cdot P_i} P_i$$

$$P_2 = u_2 - \frac{u_2 \cdot P_1}{P_1 \cdot P_1} \cdot P_1$$

$$= \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} - \frac{3+12+0}{9+36+0} \cdot \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} - \frac{1}{3} \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} - \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$$

Now, $q_2 = \frac{P_2}{|P_2|} = \frac{1}{\sqrt{0^2+0^2+2^2}} = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$

$$= \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\therefore Q = \begin{pmatrix} q_1 & q_2 \end{pmatrix} = \begin{pmatrix} \frac{2}{\sqrt{45}} & 0 \\ 6/\sqrt{45} & 0 \\ 0 & 1 \end{pmatrix}$$

(Ans.)

\# $\overset{m \times n}{A}\, x = b$ $\quad \Big| \; A = QR$

$\Rightarrow A^T A x = A^T b$

$\Rightarrow (QR)^T QR x = (QR)^T b$

$\Rightarrow R^T Q^T Q R x = Q^T R^T b$

$\Rightarrow Q^T Q R x = Q^T b$

$\Rightarrow I R x = Q^T b$ [identity $A^T A = I$]

$$\therefore Rx = Q^T b$$

#

x	$f(x)$
-3	0
0	0
6	2

Let, $P_1(x) = a_0 + a_1 x$

So, $P_1(-3) \rightarrow a_0 - 3a_1 = 0$

$P_1(0) \rightarrow a_0 = 0$

$P_1(6) \rightarrow a_0 + 6a_1 = 2$

So, Matrix,

$$\underset{(3\times2)}{\begin{bmatrix} 1 & -3 \\ 1 & 0 \\ 1 & 6 \end{bmatrix}} \underset{(2\times1)}{\begin{bmatrix} a_0 \\ a_1 \end{bmatrix}} = \underset{(3\times1)}{\begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}}$$

So, $u_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$; $u_2 = \begin{bmatrix} -3 \\ 0 \\ 6 \end{bmatrix}$

Now for step 1; $k=1$

$$P_k = u_k - \sum_{j=1}^{k-1} \frac{u_k \cdot p_i}{p_i \, p_i} p_i$$

So, $P_1 = u_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

$$q_1 = \frac{P_1}{|P_1|} = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Now for step 2; $k=2$

$$P_k = u_k - \sum_{i=1}^{k-1} \frac{u_k \cdot p_i}{p_i \cdot p_i} p_i$$

So, $P_2 = u_2 - \sum_{j=1}^{2-1} \frac{u_2 \, p_1}{p_1 \cdot p_1} p_1$

$$= \begin{bmatrix} -3 \\ 0 \\ 6 \end{bmatrix} - \left(\frac{-3+0+6}{1+1+1} \right) \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} -3 \\ 0 \\ 6 \end{bmatrix} - \begin{bmatrix} 3 \\ 3 \\ 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -4 \\ -1 \\ 5 \end{bmatrix}$$

So, $q_2 = \frac{P_2}{|P_2|} = \frac{1}{\sqrt{(-4)^2 (-1)^2 (5)^2}} \times \begin{bmatrix} -4 \\ -1 \\ 5 \end{bmatrix}$

$$= \begin{bmatrix}$$

So, $q_2 = \begin{bmatrix} -4/\sqrt{42} \\ -1/\sqrt{42} \\ 5/\sqrt{42} \end{bmatrix}$

$q_1 = \begin{bmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}$

So, $Q = \begin{bmatrix} q_1 & q_2 \end{bmatrix}$

$= \begin{bmatrix} 1/\sqrt{3} & -4/\sqrt{42} \\ 1/\sqrt{3} & -1/\sqrt{42} \\ 1/\sqrt{3} & 5/\sqrt{42} \end{bmatrix}$

$R = \begin{bmatrix} u_1^T q_1 & u_2^T q_1 \\ 0 & u_2^T q_2 \end{bmatrix}$, $\dot{R} = \begin{bmatrix} u_1^T q_1 & u_2^T q_1 & u_3^T q_1 \\ 0 & u_2^T q_2 & u_3^T q_2 \\ 0 & 0 & u_3^T q_3 \end{bmatrix}$

On, $R = Q^T A$

$$= \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \\ -4/\sqrt{42} & -1/\sqrt{42} & 5/\sqrt{42} \end{bmatrix}_{(2\times3)} \begin{bmatrix} 1 & -3 \\ 1 & 0 \\ 1 & 6 \end{bmatrix}_{(3\times2)}$$

$$= \begin{bmatrix} \sqrt{3} & \sqrt{3} \\ 0 & \sqrt{42} \end{bmatrix}_{(2\times2)}$$

Now, $Rx = Q^T b$

So, $R_x = \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \\ -4/\sqrt{42} & -1/\sqrt{42} & 5/\sqrt{42} \end{bmatrix} \begin{bmatrix}$

$$\begin{bmatrix} \sqrt{3} & \sqrt{3} \\ 0 & \sqrt{42} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \\ -4/\sqrt{42} & -1/\sqrt{42} & 5/\sqrt{42} \end{bmatrix}_{(2\times3)} \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}_{(3\times1)}$$

$$\Rightarrow \begin{bmatrix} \sqrt{3} & \sqrt{3} \\ 0 & \sqrt{42} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 2/\sqrt{3} \\ 10/\sqrt{42} \end{bmatrix}$$

So, $\begin{bmatrix} \sqrt{3} & \sqrt{3} \\ 0 & \sqrt{42} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 2/\sqrt{3} \\ 10/\sqrt{42} \end{bmatrix}$

So, $\sqrt{42}\, a_1 = 10/\sqrt{42}$

$\therefore a_1 = \frac{5}{21}$

And, $\sqrt{3}\, a_0 + \sqrt{3}\, a_1 = \frac{2}{\sqrt{3}}$

$\Rightarrow \therefore a_0 = \frac{3}{7}$

So, the equation will be;

$\frac{3}{7} + \frac{5}{21} x$ *

$f(1) = 2$; $f(2) = 3$; $f(3) = 6$; $f(4) = 4$

Let,

$$P_2(x) \rightarrow a_0 + a_1 x + a_2 x^2 = 0$$

$P_2(1) \rightarrow a_0 + a_1 + a_2 = 2$

$P_2(2) \rightarrow a_0 + 2a_1 + 4a_2 = 3$

$P_2(3) \rightarrow a_0 + 3a_1 + 9a_2 = 6$

$P_2(4) \rightarrow a_0 + 4a_1 + 16a_2 = 4$

So, the matrix will be,

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \\ 1 & 4 & 16 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 6 \\ 4 \end{bmatrix}$$

(4×3) (3×1) (4×1)

Now, $Ax = b$

$\Rightarrow$ Now, we know,

$Q = (a_1 \quad a_2 \quad a_3)$; $R = Q^T A$,

$Rx = Q^T b$

So, $u_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$ $u_2 = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$ $u_3 = \begin{bmatrix} 1 \\ 4 \\ 9 \\ 16 \end{bmatrix}$

for step-1 ; k=1

$P_1 = u_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$

$q_1 = \frac{P_1}{|P_1|} = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$

for step-2 ; k=2

$$P_2 = u_2 - \sum_{i=1}^{2-1} \frac{u_2 P_1}{P_1 P_1} \cdot P_1$$

$$= \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} - \left[\frac{10}{4} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \right]$$

$$= \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} - \begin{bmatrix} 10/4 \\ 10/4 \\ 10/4 \\ 10/4 \end{bmatrix}$$

$$= \begin{bmatrix} -1.5 \\ -0.5 \\ 0.5 \\ 1.5 \end{bmatrix}$$

$q_2 = -$

for step

$P_3 = u_3$

$= \begin{bmatrix} 1 \\ 4 \\ 9 \\ 16 \end{bmatrix}$

$= \begin{bmatrix} 1 \\ 4 \\ 9 \\ 16 \end{bmatrix}$

$=$

$$q_2 = \frac{P_2}{|P_2|} = \frac{1}{\sqrt{5}} \begin{bmatrix} -1.5 \\ -0.5 \\ 0.5 \\ 1.5 \end{bmatrix}$$

for step-3; k=3

$$P_3 = u_3 - \frac{u_3 P_1}{P_1 P_1} P_1 - \frac{u_3 P_2}{P_2 \cdot P_2} P_2$$

$$= \begin{bmatrix} 1 \\ 4 \\ 9 \\ 16 \end{bmatrix} - \left[\frac{30}{4} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \right] - \left[\frac{25}{5} \begin{bmatrix} -1.5 \\ -0.5 \\ 0.5 \\ 1.5 \end{bmatrix} \right]$$

$$= \begin{bmatrix} 1 \\ 4 \\ 9 \\ 16 \end{bmatrix} - \begin{bmatrix} 7.5 \\ 7.5 \\ 7.5 \\ 7.5 \end{bmatrix} - \begin{bmatrix} -3.75 \\ -1.25 \\ 1.25 \\ 3.75 \end{bmatrix}$$

$$= \begin{bmatrix} 2.875 \\ \vdots \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 4 \\ 9 \\ 16 \end{bmatrix} - \begin{bmatrix} 7.5 \\ 7.5 \\ 7.5 \\ 7.5 \end{bmatrix} - \begin{bmatrix} -15/2 \\ -5/2 \\ 5/2 \\ 15/2 \end{bmatrix}$$

$=$

Using another formula,

$$u_3 - \{(u_3^T a_1) a_1 + (u_3^T a_2) a_2\}$$

$$u_3^T a_1 = \begin{pmatrix} 1 & 4 & 9 & 16 \end{pmatrix} \begin{bmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{bmatrix}$$

$$= 15 \times a_1$$

$$= 15 \times \begin{bmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{bmatrix}$$

$$= \begin{bmatrix} 7.5 \\ 7.5 \\ 7.5 \\ 7.5 \end{bmatrix}$$

Now,

$$(u_3^T a_2)\, a_2$$

$$\text{so, } u_3^T a_2 = \begin{bmatrix} 1 & 4 & 9 & 16 \end{bmatrix} \begin{bmatrix} -1.5/\sqrt{5} \\ -0.5/\sqrt{5} \\ 0.5/\sqrt{5} \\ 1.5/\sqrt{5} \end{bmatrix}$$

$$= 5\sqrt{5} \times a_2$$

$$= 5\sqrt{5} \begin{bmatrix} -1.5/\sqrt{5} \\ -0.5/\sqrt{5} \\ 0.5/\sqrt{5} \\ 1.5/\sqrt{5} \end{bmatrix} = \begin{bmatrix} -15/2 \\ -5/2 \\ 5/2 \\ 15/2 \end{bmatrix}$$

$$\text{So, } p_3 = \begin{bmatrix} 1 \\ 4 \\ 9 \\ 16 \end{bmatrix} - \begin{bmatrix} 7.5 \\ 7.5 \\ 7.5 \\ 7.5 \end{bmatrix} - \begin{bmatrix} -15/2 \\ -5/2 \\ 5/2 \\ 15/2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix}$$

$$q_3 = \frac{1}{2} \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix}$$

So, $Q = (a_1 \quad a_2 \quad a_3)$

$$= \begin{bmatrix} 0.5 & -1.5/\sqrt{5} & 0.5 \\ 0.5 & -0.5/\sqrt{5} & -0.5 \\ 0.5 & 0.5/\sqrt{5} & -0.5 \\ 0.5 & 1.5/\sqrt{5} & 0.5 \end{bmatrix}$$

Now,

$$R = Q^T A$$

$$= \begin{bmatrix} 0.5 & 0.5 & 0.5 & 0.5 \\ -1.5/\sqrt{5} & -0.5/\sqrt{5} & 0.5/\sqrt{5} & 1.5/\sqrt{5} \\ 0.5 & -0.5 & -0.5 & 0.5 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \\ 1 & 4 & 16 \end{bmatrix}$$

$=$

$$R = \begin{bmatrix} u_1^T a_1 & u_2^T a_1 & u_3^T a_1 \\ 0 & u_2^T a_2 & u_3^T a_2 \\ 0 & 0 & u_3^T a_3 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 5 & 15 \\ 0 & \sqrt{5} & 5\sqrt{5} \\ 0 & 0 & 2 \end{bmatrix}$$

$$R_x = Q^T b$$

$$\begin{bmatrix} 2 & 5 & 15 \\ 0 & \sqrt{5} & 5\sqrt{5} \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} \cdot \\ \cdot \\ \cdot \end{bmatrix}$$

$$\begin{bmatrix} 2 \\ 3 \\ 6 \\ 4 \end{bmatrix}$$